

GOKUL RAVICHANDRAN

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Professional Summary

Machine Learning Engineer with 4+ years of experience developing and deploying production-grade AI and machine learning solutions across healthcare, cloud, and industrial automation domains. Specialized in Generative AI, NLP, Computer Vision, and LLMs, with expertise in building scalable, high-performance systems that bridge research innovation and real-world applications. Proficient in Python, TensorFlow, PyTorch, and LangChain, with strong hands-on experience in data engineering, MLOps, and model optimization. Adept at leveraging cloud platforms (AWS, Azure, GCP) to deliver robust, secure, and efficient AI applications, consistently improving accuracy, reliability, and business impact across enterprise environments.

Technical Skills

Programming & Scripting: Python, R, SQL, Bash, JavaScript
Machine Learning & Deep Learning: Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, LSTM, CNN, Transformers (BERT, GPT, T5, LLaMA)
Generative AI & LLMs: LangChain, LlamaIndex, GPT-4, Gemini, RAG Pipelines, Prompt Engineering, Reinforcement Learning (PPO, DDPG), RLHF, Bandit Algorithms
Data Engineering & Processing: Apache Spark, PySpark, Snowflake, Apache NiFi, AWS Glue, Airflow, DVC, SQL, Pandas, NumPy, ChromaDB, FAISS, Pinecone
MLOps & Deployment: Docker, Kubernetes, FastAPI, MLflow, GitHub Actions, Azure Kubernetes Service (AKS), SageMaker, Vertex AI, CI/CD, Model Versioning, Monitoring

Computer Vision & NLP: YOLOv5/v8, OpenCV, FastAI, spaCy, NLTK, OCR, Vision-Language Models, Text Summarization, Sentiment Analysis
Cloud Platforms: AWS (S3, Lambda, SageMaker), Azure (Cognitive Search, Databricks, AKS), GCP (Vertex AI, BigQuery)
Databases & Storage: Snowflake, Redshift, MongoDB, PostgreSQL, Elasticsearch, ChromaDB
Visualization & Analytics: Power BI, Tableau, Grafana, Streamlit, Gradio
Tools & Dev Practices: Git, JIRA, Agile, Weights & Biases (W&B), Grad-CAM, SHAP, QGIS, EconML, DoWhy

Professional Experience

AI/ML Engineer | Cardinal Health, USA Feb 2025 - Present

- Developed a GenAI-powered clinical assistant (ReX-Health) using **GPT-4, Gemini, and LangChain**, integrating **RAG, Pinecone, and LlamaIndex** to improve **clinical query accuracy by 35%** and increase **physician adoption by 30%**.
- Engineered scalable vector search pipelines with **FAISS, ChromaDB, and OpenAI embeddings**, enabling **sub-second retrieval** of **EHR and medical knowledge base** records across 50K+ patient documents.
- Built automated CI/CD workflows using **Docker, MLflow, and GitHub Actions**, reducing model deployment time from hours to minutes and ensuring **audit-ready healthcare compliance**.
- Containerized multi-LLM microservices via **FastAPI and Kubernetes**, supporting **autoscaling clinical inference endpoints** across **Azure Kubernetes Service (AKS)** for large-scale healthcare workloads.
- Deployed an end-to-end model monitoring system using **Prometheus and Grafana**, tracking model drift on **diagnostic models** and reducing **clinical prediction errors by 22%**.
- Integrated Azure Cognitive Search for hybrid **semantic + keyword retrieval** of drug, procedure, and patient literature, boosting **clinical query accuracy by 41%**.
- Designed RLHF-style feedback loops using **clinician validation, prompt ranking, and Bandit algorithms**, enhancing model reliability and improving **AI-driven decision support** for healthcare professionals.

Machine Learning Engineer | Hellinex Cloud, India Dec 2020 - Jul 2023

- Developed digital twin simulation models using **Reinforcement Learning (PPO, DDPG)** in **PyTorch**, optimizing irrigation and fertilizer scheduling strategies that **increased simulated crop yield accuracy by 18%**.
- Built real-time soil anomaly detection systems utilizing **YOLOv5, OpenCV, and geospatial QGIS overlays**, delivering **92%+ detection precision** and enabling **data-driven agronomic intervention** across monitored regions.
- Engineered scalable data ingestion pipelines using **Snowflake and Apache Spark** to automate the processing of satellite and IoT sensor data, enabling **faster analysis** and improving the **efficiency of multi-source data workflows**.
- Researched and authored an internal **whitepaper on Causal Inference in Deep Learning** employing **DoWhy, EconML, and SHAP**, enhancing **model explainability and stakeholder confidence** in ML-driven predictions.
- Delivered interactive Power BI dashboards integrated with **prescriptive AI insights**, empowering **cross-functional teams** to make **real-time data-backed decisions** and improving **operational response efficiency by 25%**.

Machine Learning Engineer | Parallax Pros, India Jun 2019 - Nov 2020

- Developed and deployed CNN-powered visual defect detection systems using **TensorFlow and TFLite** on **edge devices**, enhancing **industrial IoT quality inspection** throughput by **40%** across automated production lines.
- Implemented a real-time **fraud detection engine** leveraging **Isolation Forests, Autoencoders, and Z-score analytics**, reducing **false positives by 33%** and improving **transactional reliability** across enterprise systems.
- Built an LSTM-based predictive maintenance framework on **Azure Databricks**, forecasting equipment failures with **28% higher accuracy**, minimizing **unplanned downtime**, and optimizing **maintenance scheduling**.
- Automated complex ETL workflows using **Apache Airflow and DVC**, cutting **data pipeline latency by 45%** and enabling **scalable, version-controlled model retraining** across distributed data environments.
- Optimized data storage and retrieval in **MongoDB and PostgreSQL** via **vector-based similarity search**, improving **dashboard response time** from **5 seconds to under 1.2 seconds** for real-time analytics.

Education

Masters of Science in Computer Science Aug 2023 - May 2025

Indiana State University, Terre Haute, IN

Bachelors of Engineering in Computer Science Jun 2016 - Aug 2020

Anna University, Chennai, India

Projects

SmartCrop Vision – Precision Agriculture Image Diagnosis

- Developed a precision agriculture system using **CNNs, FastAI, FastAPI, and Docker** to identify crop diseases from field images; integrated **MongoDB and Power BI dashboards** for real-time visualization, delivering **92% classification accuracy** and reducing manual inspection time by **40%** through Agile sprints in **JIRA**.

NewsSense – Real-Time News Summarization and Sentiment Pipeline

- Engineered a real-time NLP workflow using **T5, BART, and Pegasus** from **Hugging Face Transformers**, combined with **spaCy, NLTK, and custom regex** for text preprocessing; deployed via **FastAPI and Kafka** to stream news summaries and sentiment alerts, reducing **information processing time by 60%** for research teams.